Who Covers Whom? Word Coverage Matrices for Human Text and for LLMs of the Same and Different Families

Anonymous ACL submission

Abstract

The Word Coverage Score (WCS) measures whether the words of a text stay inside the top-p nucleus of a scoring model. We use it as an instrument on a scorer-by-source matrix in which human text is one more generator next to LLMs, so that human versus LLM, same-family LLM pairs and cross-family LLM pairs are read from one object. A first run (50 questions, 7 LLM sources, 7 scorers, two families) gives four observations. First, human text has the lowest coverage under every scorer (0.35–0.78 against 0.75–1.00 for LLM text). Second, the gap is consistent: 97 of 98 cells have a 95% interval below zero. Third, among LLMs the scorer family matters more than the source family, and a same-family advantage is visible but small. Fourth, absolute values are scorer-relative. We then specify the experiments needed to test these readings across datasets, domains, families and sizes; their results are marked [XX].

1 Introduction

The Word Coverage Score (WCS) (Anonymous, 2026) asks, at each position of a text, whether the observed word lies inside the top-p nucleus of a scorer model, and averages over the text. It was proposed to study which words remain reachable under sampling. Because any model can act as scorer and any text as source, WCS gives a *scorer-by-source matrix*.

We put human text in that matrix as one more generator. This lets one object answer three related questions:

Q1. **Human vs. LLM.** Where does human text sit among LLM sources, for each scorer?

Q2. **Same family.** How does coverage change between LLMs of one family (different sizes, base vs. instruct, different versions)?

Q3. **Different families.** How does coverage change when scorer and source come from different families, and is it symmetric?

The direction of Q1 is not new: zero-shot detectors exploit the fact that human text is less predictable to language models (Gehrmann et al., 2019; Hans et al., 2024). We do not claim a detector. Our reading is that WCS is an instrument, and the matrix is the tool that puts Q1–Q3 on one scale.

Contributions: (1) a matrix protocol that treats human text as a generator and defines contrasts for human–LLM, same-family and cross-family cells, with paired bootstrap intervals; (2) a first matrix on 50 questions, 7 LLMs and 7 scorers; (3) a concrete experiment plan with datasets, models, sample sizes and compute (Section 5, Appendices A–C).

2 Related Work

Closest same-recipe work. Two recent papers also look for a statistic that separates human from machine text without training a classifier. nDMD (Wang et al., 2026) is a locality-based statistic that is higher in human text, evaluated on M4 and a detection shared task (Findings of EACL 2026). Telescope (Nassif and Cooper, 2026) uses token-repetition probability for zero-shot detection. Both measure a human–LLM gap with a new statistic and evaluate detection. We evaluate neither detection nor a threshold; we use WCS to compare sources, including LLM against LLM.

Zero-shot detection. GLTR (Gehrmann et al., 2019) showed that machine text concentrates in high-probability ranks. DetectLLM (Su et al., 2023) and Binoculars (Hans et al., 2024) turn rank and perplexity information into detectors. Koike et al. (2025) show that detection and membership-inference methods approximate one optimal metric, which places our scorers in the same family of statistics.

Between LLMs. Artificial Hivemind (Jiang et al., 2025) reports that different LLMs give strikingly similar answers to open-ended prompts, which makes inter-model comparison a question about shared distributions. WCS asks it per word and per scorer, with a direction.

3 Method

Matrix. Let S be the set of scorer models and G the set of sources, where G contains the LLMs and one human source. For prompt q, scorer s and source g, $\mathrm{WCS}_{s,g}(q)$ is the fraction of words of the text $g(q)$ that lie in the top-p nucleus of s under teacher forcing (Anonymous, 2026). We store the top-100 probabilities per position, so variants (p, strict or renormalized nucleus) are computed offline. The cell is the mean over prompts.

Relation classes. Each cell (s, g) belongs to one class: *human* (g human), *self* ($g = s$), *same family, other size*, *same family, other stage* (base vs. instruct), *same family, other version*, and *cross family*. Because the scorer and the source play different roles, classes are kept directional: s covering g is not g covering s.

Contrasts. For two classes A and B we report $\Delta = \overline{\mathrm{WCS}}_A - \overline{\mathrm{WCS}}_B$ with a 95% paired bootstrap interval, resampling prompts (the same prompts underlie every cell). Human versus LLM uses the per-prompt difference $d = \mathrm{WCS}(\mathrm{human}) - \mathrm{WCS}(\mathrm{LLM})$ per scorer and LLM source.

4 First Run

Setup. Data: 50 English ELI5 questions from HC3 (Guo et al., 2023) (seed 42) with the curated human answers (150–400 words, written before 2022). Sources: human plus 7 LLMs answering the same questions with greedy decoding: Qwen3.5-2B, 4B, 9B, 27B (int8) (Qwen Team, 2026) and Gemma-4-E2B-it, E4B-it, 12B-it (Hugging Face, 2026). Scorers: the same 7 models. Metric: top-p 0.99 with the stored top-100 renormalized (realistic); the raw-nucleus (strict) variant is in Appendix D. A greedy-argmax variant was unstable across inference engines (mean absolute difference 0.159 on the same texts) and is excluded. The int8 scorer differs from its bf16 version by 0.020 (realistic) on the human texts, inside the 0.015–0.089 band between bf16 engines.

Q1: human vs. LLM. Human text has the lowest coverage in every column of Figure 1:

Scorer	Human	LLM range	Min d [95% CI]
Qwen-2B	0.68	0.96–1.00	-0.28 $[-0.34, -0.22]$
Qwen-4B	0.74	0.98–1.00	-0.23 $[-0.29, -0.18]$
Qwen-9B	0.78	0.97–1.00	-0.19 $[-0.25, -0.14]$
Qwen-27B*	0.77	0.94–1.00	-0.16 $[-0.22, -0.10]$
Gem-E2B	0.35	0.75–1.00	-0.40 $[-0.48, -0.31]$
Gem-E4B	0.53	0.85–1.00	-0.31 $[-0.39, -0.23]$
Gem-12B	0.67	0.80–1.00	-0.12 $[-0.20, -0.04]$

Table 1: Realistic top-p 0.99 WCS per scorer: human mean, range over the 7 LLM sources, and the least negative paired difference d (human − LLM). *int8.

	Qwen source	Gemma source
Qwen scorers (4)	0.99	0.97
Gemma scorers (3)	0.84	0.88

Table 2: Mean WCS over LLM sources by family of scorer and source (realistic top-p 0.99, rounded cell values of Fig. 1, diagonal excluded). Descriptive; no CIs yet.

0.35 (Gemma-4-E2B scorer) to 0.78 (Qwen3.5-9B scorer), while LLM rows stay in 0.75–1.00. The difference d is negative in all 98 cells (7 scorers $\times$ 7 sources $\times$ 2 metrics), with the 95% interval fully below zero in 97. The only boundary cell is Gemma-4-12B-it scoring Qwen3.5-2B text, strict metric: $d = -0.08$, CI $[-0.151, +0.001]$. Table 1 gives the least negative d per scorer.

Q2 and Q3: LLM against LLM. Reading the LLM block of Figure 1 (means of the rounded cell values, off-diagonal, no intervals yet; intervals [XX]) gives the pattern in Table 2. Qwen scorers cover all LLM text almost fully (0.99 on Qwen text, 0.97 on Gemma text). Gemma scorers are lower on everything, and lowest for the smallest scorer. The same-family advantage is small: Qwen scorers drop 0.01 from Qwen to Gemma text, Gemma scorers drop 0.04 from Gemma to Qwen text. Within a family, size is not a clear factor in this run: Qwen scorers saturate, and for Gemma the E2B scorer is the strictest. The scorer family explains more of the spread than the source family. Cells are not symmetric: Gemma scorers cover Qwen text less than Qwen scorers cover Gemma text (0.84 vs. 0.97). An earlier internal run on other prompts, with more models, points the same way: the same-family mean was 0.94 and the cross-family mean 0.90 in that run [XX] (metric and CIs to be confirmed).

Controls. Topic is fixed because all sources answer the same questions. Length is nearly uncor-

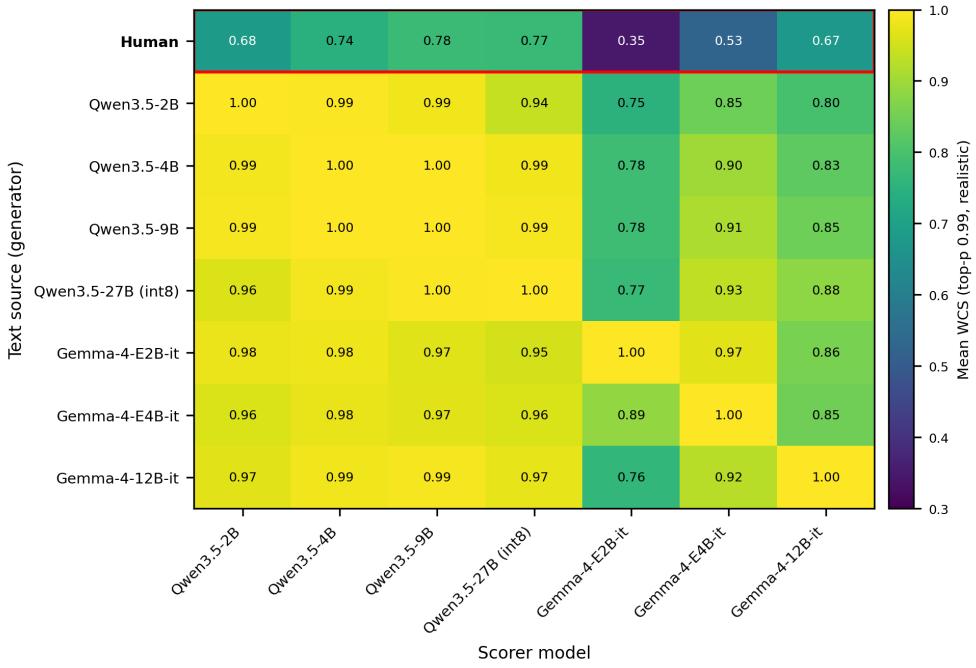


Figure 1: Mean WCS (top-p 0.99, realistic) for each scorer (columns) and each source (rows). Human text is the top row (red box), one more generator next to the 7 LLMs. LLM rows are greedy generations; Qwen3.5-27B is int8.

related with WCS here ($|r| \approx 0.05$). Quantization is checked above. The human answers predate 2022 and are probably in every scorer's pretraining data; memorization would raise human coverage, so contamination makes the gap conservative. The same scorers give 0.84–0.99 LLM-to-LLM coverage on an independent prompt set, so the low human row is not a scorer artifact.

5 Proposed Experiments

The first run has one dataset, one register, two families and 50 prompts. Table 3 lists the experiments needed to answer Q1–Q3; models, datasets and compute are detailed in Appendices A–C. Results not yet obtained are [XX].

What each shows. E1 tests whether Q1 is a property of ELI5 forum text or holds in news, reviews, encyclopedic and scientific writing. E2 separates a size effect from a stage effect (base vs. instruct) with the same recipe. E3 tests the direction seen in Table 2 with a third, fully open family. E4 attacks the main confounds: the human–LLM gap may partly be a register gap, and greedy text is the most predictable case.

Analysis. All contrasts are paired over prompts with 95% bootstrap intervals, reported per dataset and pooled. We report the human–LLM Δ and the same-family minus cross-family Δ as the two headline numbers.

Future work. The matrix makes a mechanism study possible: split coverage into words inside and outside the intersection of two models' nuclei, and generate from the intersection and from the complement. This is planned work and no mechanism result is reported here.

Limitations

(1) The measured results use 50 questions, one dataset and one domain. (2) Human and LLM text differ in register, not only in origin. (3) LLM rows are greedy only. (4) The 27B model is int8; quantization was checked but a residual bias is possible. (5) Two families in the measured run. (6) The LLM-to-LLM reading in Table 2 uses rounded cell values and has no intervals. (7) Pretraining contamination of old human text affects absolute values. (8) No detection claim: no threshold, AUROC or adversarial test.

References

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	Question	Design	n and compute	Result
E1	Does the human–LLM gap hold across datasets, domains and families? (Q1)	Human text plus 8 LLMs (3 families) on 4 human-text datasets (HC3, M4, RAID, MAGE); 8 scorers; Δ per domain	250 prompts per dataset; $\approx$5 H100-h	[XX]
E2	Does size, training stage or version inside a family change coverage? (Q2)	Qwen3.5 (6 sizes) and Gemma 4 (5 sizes), base and instruct; full 22×22 matrix; Δ for each relation class	500 prompts; $\approx$10–15 H100-h	[XX]
E3	Is cross-family coverage lower and directional? (Q3)	Qwen3.5, Gemma 4, Olmo 3 at 3 sizes each; 9×9 matrix on E1 generations; Infinity-Chat prompts as open-ended domain	1000 prompts; reuses E1	[XX]
E4	Do register, decoding and recency explain the gap? (Q1)	Sampled generations (not only greedy); LLM text in forum register; human text after the models' cutoffs	250 prompts; $\approx$5 H100-h	[XX]

Table 3: Experiment plan. Compute is a rough estimate from token counts and assumed throughput (Appendix C).

Liam Dugan, Alyssa Hwang, Filip Trhlik, Josh Magnus Ludan, Andrew Zhu, Hainiu Xu, Daphne Ippolito, and Chris Callison-Burch. 2024. RAID: A shared benchmark for robust evaluation of machine-generated text detectors. In *Proceedings of ACL 2024 (Long Papers)*.

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A Models

All models are open weights. Hardware constraint: every model used in the plan fits on a single 94 GB GPU at int8 precision or higher, weights plus

KV-cache headroom included (the largest planned checkpoints, Qwen3.5-35B-A3B, Gemma-4-31B and Olmo-3-32B, take roughly 31–64 GB). Checkpoints are listed from their release pages; exact checkpoint ids are [XX] until fixed in the run configuration.

- **Qwen3.5** (Qwen Team, 2026): sizes from 0.8B to 35B-A3B in the collection (2B, 4B, 9B, 27B already used). Larger members (122B-A10B, 397B-A17B) exceed the 94 GB single-GPU budget even at int8 and are excluded.
- **Gemma 4** (Hugging Face, 2026): E2B, E4B, 12B, 26B-A4B (MoE), 31B; base and instruction-tuned checkpoints for each size.
- **Olmo 3 / 3.1** (Team Olmo, 2025): 7B and 32B, fully open data and training flow, which allows checking contamination. Instruct, Think and base checkpoints; Olmo 3.1 32B Instruct is listed on the Ai2 release page (https://allenai.org/blog/olmo3).
- **Excluded (do not fit 94 GB at int8):** Mistral Small 4 (Mistral AI, 2026) (119B total, ≈119 GB at int8) and the Qwen3.5 122B-A10B and 397B-A17B checkpoints. They would need multi-GPU or sub-8-bit quantization and stay out of the plan.

Core LLM sources for E1/E3: Qwen3.5 {2B, 9B, 27B}, Gemma 4 {E4B, 12B, 31B}, Olmo 3 {7B, 32B}. Dense models up to about 32B are run in bf16 or int8 as in the first run; the 35B-A3B MoE runs at int8. All of these stay within the 94 GB budget.

B Datasets

We use datasets for human text and prompts, and regenerate LLM text with our own models, so that all sources answer the same prompts and recent models are used.

- **HC3** (Guo et al., 2023): human expert and ChatGPT answers to open-domain, finance, medicine, law and psychology questions. Used so far for ELI5 only; the other domains are new. https://arxiv.org/abs/2301.07597
- **M4** (Wang et al., 2024): multi-generator, multi-domain, multi-lingual corpus; gives human text in several domains and, if justified, non-English text. Domain and language list to be taken from the dataset card [XX]. https://aclanthology.org/2024.eacl-long.83
- **RAID** (Dugan et al., 2024): over 6 million generations, 8 domains, 11 models; we use its human text and prompts. https://aclanthology.org/2024.acl-long.674
- **MAGE** (Li et al., 2024): human writings from diverse sources. https://aclanthology.org/2024.acl-long.3
- **Infinity-Chat** (Jiang et al., 2025): 26K open-ended user queries without human answers; used for LLM-to-LLM matrices (E3) as in our earlier internal run. https://arxiv.org/abs/2510.22954

Domain is a factor in every analysis (QA, encyclopedic, news, reviews, scientific abstracts, open-ended chat). Whether the license of each human corpus allows our use is [XX].

C Compute

Estimates only, from token counts and assumed throughput that we have not measured: texts of about 400 tokens; teacher-forced scoring at about 10k tokens/s on one H100 (bf16, mixed sizes); generation at about 2k tokens/s per model with batching. E1: 9 sources × 1000 prompts × 8 scorers is about 29M scored tokens (≈1 h) plus 3.2M generated tokens per model set (≈3 h). E2: 22 sources and scorers, 500 prompts, about 97M scored tokens (≈3–4 h) plus generation (≈3 h). E3 reuses E1. E4 adds sampled and register-matched generations. All runs are single-GPU: every model in the plan fits one 94 GB H100 at int8 or better (Appendix A). Total ≈25–35 H100-hours, to be replaced by measured values. Storing the top-100 probabilities per position takes roughly 0.25 MB per text and scorer, so E2 needs on the order of 60 GB.

D Strict-metric matrix

Figure 2 repeats Figure 1 with the strict top-p 0.99 variant (raw nucleus, no renormalization). The human row is again the lowest under every scorer.

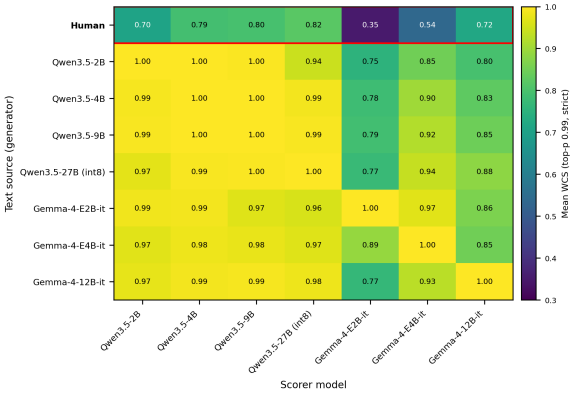


Figure 2: Mean WCS (top-p 0.99, strict) by scorer and source; human text in the top row.

Scorer	Human	LLM range	Min d [95% CI]
Qwen-2B	0.70	0.97–1.00	-0.26 $[-0.32, -0.21]$
Qwen-4B	0.79	0.98–1.00	-0.19 $[-0.25, -0.14]$
Qwen-9B	0.80	0.97–1.00	-0.17 $[-0.23, -0.12]$
Qwen-27B*	0.82	0.94–1.00	-0.12 $[-0.18, -0.06]$
Gem-E2B	0.35	0.75–1.00	-0.39 $[-0.48, -0.31]$
Gem-E4B	0.54	0.85–1.00	-0.30 $[-0.38, -0.22]$
Gem-12B	0.72	0.80–1.00	-0.08 $[-0.15, 0.00]$

Table 4: Strict top-p 0.99 WCS per scorer, same layout as Table 1. *int8.